Soal No 1 (File Singlylist.h)

```
#ifndef SINGLYLIST_H
#define SINGLYLIST_H

typedef int infotype;
typedef struct ElmList *address;

struct ElmList {
   infotype info;
   address next;
};

struct List {
   address First;
};

void createList(List &L);
address alokasi(infotype x);
void dealokasi(address &P);
void printInfo(List L);
void insertFirst(List &L, address P);
address findElem(List L, infotype x);
int totalInfo(List L);

#endif
```

Soal No 2 (File Singlylist.cpp)

```
#include <iostream>
#include "Singlylist.h"
using namespace std;

void createList(List &L) {
   L.First = NULL;
}

address alokasi(infotype x) {
   address P = new ElmList;
   P->info = x;
   P->next = NULL;
   return P;
}
```

```
void dealokasi(address &P) {
   delete P;
   P = NULL;
}

void printInfo(List L) {
   address P = L.First;

   while (P != NULL) {
      cout << P->info << " ";
      P = P->next;
   }
   cout << endl;
}

void insertFirst(List &L, address P) {
   P->next = L.First;
   L.First = P;
}

address findElem(List L, infotype x) {
   address P = L.First;

   while (P != NULL) {
      if (P->info == x) {
         return P;
      }
      P = P->next;
   }

   return NULL;
}

int totalInfo(List L) {
   address P = L.First;
   int total = 0;

   while (P != NULL) {
      total += P->info;
      P = P->next;
   }

   return total;
```

```
}
```

Soal No 3 (File main.cpp)

```
#include <iostream>
#include "Singlylist.h"
using namespace std;

int main() {
   List L;
   address P1, P2, P3, P4, P5 = NULL;

   createList(L);

   P1 = alokasi(2);
   P2 = alokasi(0);
   P3 = alokasi(8);
   P4 = alokasi(12);
   P5 = alokasi(9);

   insertFirst(L, P1);
   insertFirst(L, P2);
   insertFirst(L, P3);
   insertFirst(L, P4);
   insertFirst(L, P5);

   cout << "Isi list: ";
   printInfo(L);

   address P = findElem(L, 8);

   if (P != NULL) {
      cout << "8 ditemukan dalam list" << endl;
   } else {
      cout << "8 tidak ditemukan dalam list" << endl;
   }

   cout << "Total info dari kelima elemen adalah "
        << totalInfo(L) << endl;

   return 0;
}
```

Soal No 4 (Output program)

Hasil yang benar

Isi list: 9 12 8 0 2

8 ditemukan dalam list

Total info dari kelima elemen adalah 31